# Lab Report: Image classification experiments

## 1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. The learning rate was set to 0.001 with the Adam optimizer.

## 2 Results

Figure 1 shows the training loss.

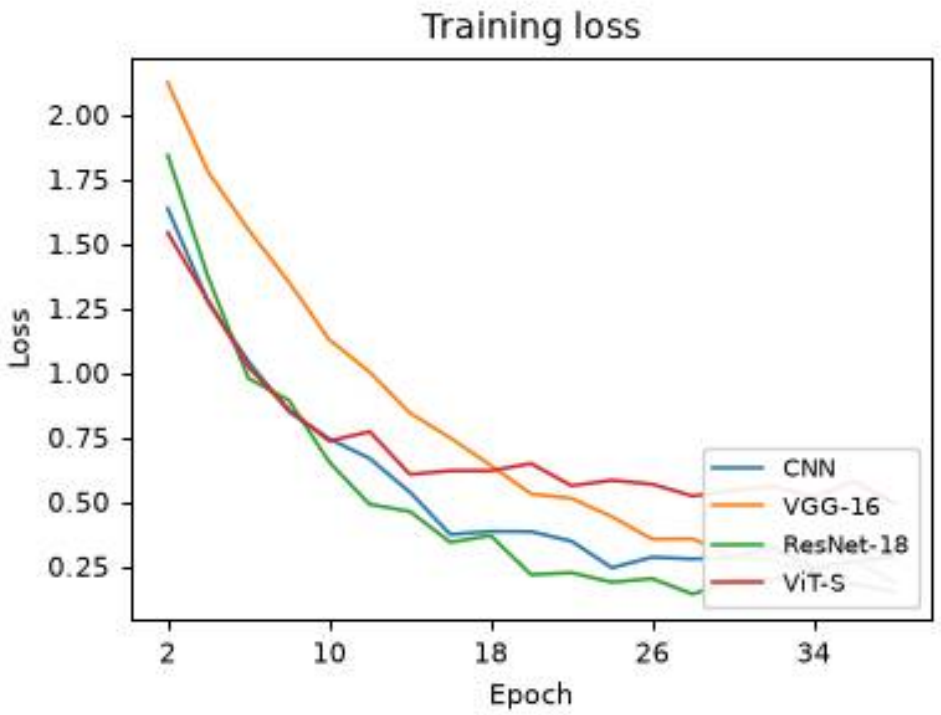



Figure 1: Training loss for the models in this experiment.

At epoch 10, the loss of ViT-S is 0.734. CNN has a loss of 0.667 at epoch 12 (Figure 1). As shown in Figure 1, at epoch 8, the loss of CNN is about 0.8. VGG-16 has a loss of 1.182 at epoch 16.

Figure 2 shows the f1 score of DistilBERT by dataset.

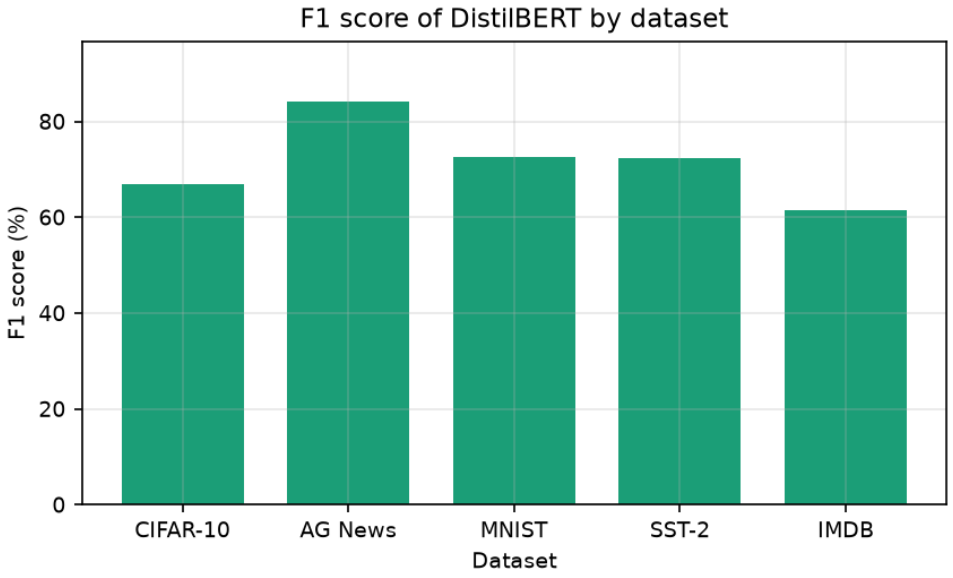



Figure 2: F1 score of DistilBERT by dataset for the models in this experiment.

DistilBERT achieves 72.3% F1 score on SST-2. On CIFAR-10, DistilBERT obtains a F1 score of 84.1% (Figure 2).

Figure 3 shows the test accuracy during training.

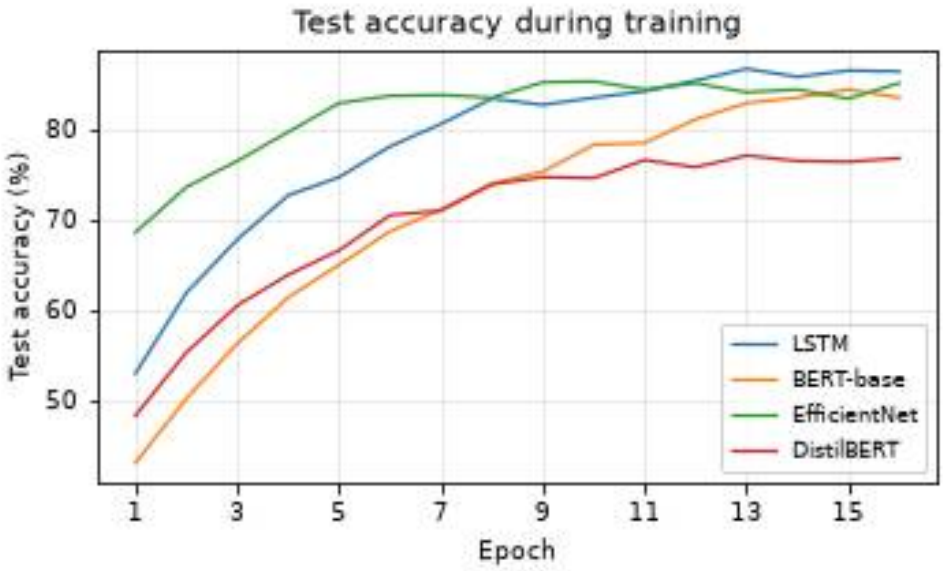



Figure 3: Test accuracy during training for the models in this experiment.

From epoch 2 to epoch 6, the test accuracy of LSTM improves by a relative 27.8% (Figure 3). LSTM attains a test accuracy of 83.3% at epoch 8 (Figure 3). LSTM reaches 61.3% test accuracy at epoch 12. LSTM attains a test accuracy of 78.0% at epoch 6 (Figure 3).

## 3 Discussion

Logistic Regression reaches 58.5% accuracy on Fashion-MNIST. The differences between the models are discussed in the next section.